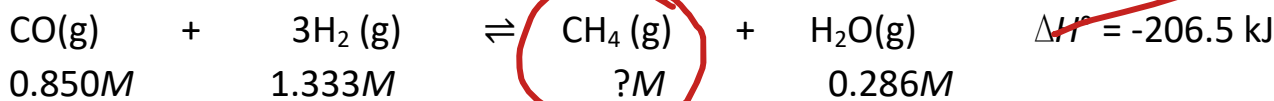


Chemical Equilibrium

only Lesson 3 K_{eq} , $[]$ equilibrium Constant

Section 3: Calculating Equilibrium Concentrations



Calculate the concentration of CH_4 in the equilibrium mixture if $K_{eq} = 3.933$.

$$K_{eq} = \frac{[\text{CH}_4][\text{H}_2\text{O}]}{[\text{CO}][\text{H}_2]^3} \quad 3.933 = \frac{[\text{CH}_4](0.286)}{(0.850)(1.333)^3} \Rightarrow [\text{CH}_4] = 27.68 \text{ M}$$

At 1405 K, The equilibrium constant for the reaction is 2.27×10^{-3} . $\rightarrow K_{eq}$



What is the concentration of hydrogen gas if $[\text{S}_2] = 0.0540 \text{ mol/L}$ and $[\text{H}_2\text{S}] = 0.184 \text{ mol/L}$?

$$K_{eq} = \frac{[\text{H}_2]^2[\text{S}_2]}{[\text{H}_2\text{S}]^2} \quad 2.27 \times 10^{-3} = \frac{[\text{H}_2]^2(0.0540)}{(0.184)^2} \Rightarrow [\text{H}_2] = 1.423 \text{ mol/L}$$

At a certain temperature, $K_{eq} = 10.5$ for the equilibrium $\text{CO(g)} + 2\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_3\text{OH(g)}$.

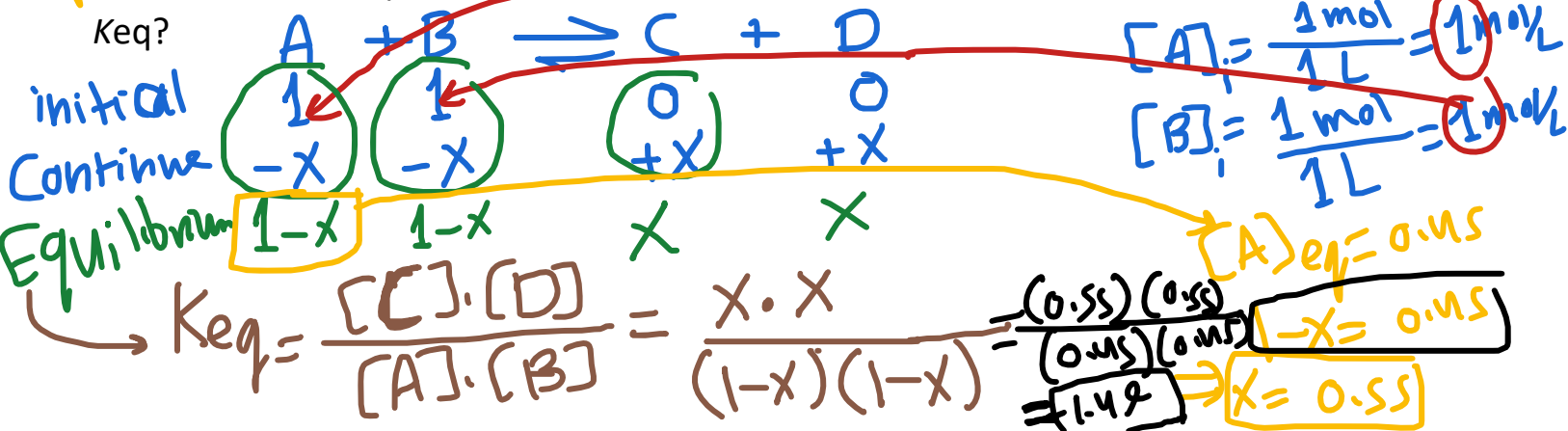
Calculate $[\text{CO}]$ in an equilibrium mixture containing 0.933 mol/L H_2 and 1.32 mol/L CH_3OH

The reaction $\text{COCl}_2\text{(g)} \rightleftharpoons \text{CO(g)} + \text{Cl}_2\text{(g)}$ reaches equilibrium at 900 K.

K_{eq} is 8.2×10^{-2} . If the equilibrium concentrations of CO and Cl_2 are 0.150M, what is the equilibrium concentration of COCl_2 ?

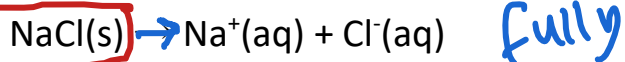
In a generic reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$, 1.00 mol of A and 1.00 mol of B are allowed to react in a 1L flask until equilibrium is established. If the equilibrium concentration of A is 0.450

mol/L, what is the equilibrium concentration of each of the other substances? What is K_{eq} ?



The Solubility Product Constant

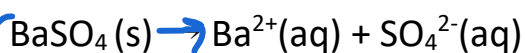
Some ionic compounds such as (NaCl) dissolve readily in water, and some such as (BaSO₄) barely dissolve at all.



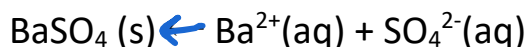
Because of the high solubility of NaCl, the oceans and some lakes contain large amounts of salt.

Sometimes low solubility is also important. Although barium ions are toxic to humans, patients must ingest barium sulfate prior to having an X ray of the digestive tract taken.

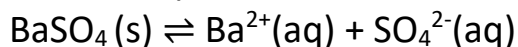
Barium sulfate dissociates in water according to this equation.



As soon as the first product ions form, the reverse reaction begins.



In time, equilibrium is established.



For sparingly soluble compounds such as BaSO₄, the rates become equal when the concentrations of the aqueous ions are exceedingly small.

Nevertheless, the solution at equilibrium is a saturated solution.

The equilibrium constant expression for the dissolving of a sparingly soluble compound is called the **solubility product constant**, *K_{sp}*.

$K_{sp} = [\text{Ba}^{2+}][\text{SO}_4^{2-}]$

The **solubility product constant expression** is the product of the concentrations of the dissolved ions, each raised to the power equal to the coefficient of the ion in the chemical equation.

Write the equation and **K_{sp}** for Mg(OH)₂



$K_{sp} = [\text{Mg}^{2+}] \cdot [\text{OH}^-]^2$

K_{eq}



Table 17.3

Solubility Product Constants at 298 K

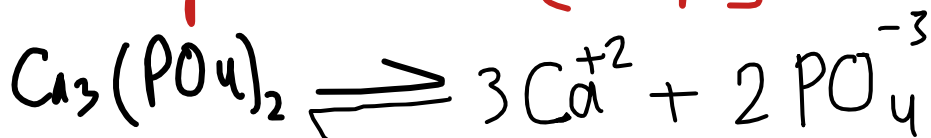
Compound	K_{sp}	Compound	K_{sp}	Compound	K_{sp}
Carbonates		Halides		Hydroxides	
BaCO ₃	2.6×10^{-9}	CaF ₂	3.5×10^{-11}	Al(OH) ₃	4.6×10^{-33}
CaCO ₃	3.4×10^{-9}	PbBr ₂	6.6×10^{-6}	Ca(OH) ₂	5.0×10^{-6}
CuCO ₃	2.5×10^{-10}	PbCl ₂	1.7×10^{-5}	Cu(OH) ₂	2.2×10^{-20}
PbCO ₃	7.4×10^{-14}	PbF ₂	3.3×10^{-8}	Fe(OH) ₃	4.9×10^{-17}
MgCO ₃	6.8×10^{-6}	PbI ₂	9.8×10^{-9}	Fe(OH) ₃	2.8×10^{-39}
Ag ₂ CO ₃	8.5×10^{-12}	AgCl	1.8×10^{-10}	Mg(OH) ₂	5.6×10^{-12}
ZnCO ₃	1.5×10^{-10}	AgBr	5.4×10^{-13}	Zn(OH) ₂	3×10^{-17}
Hg ₂ CO ₃	3.6×10^{-17}	AgI	8.5×10^{-17}	Sulfates	
Chromates		Phosphates		BaSO ₄	1.1×10^{-10}
BaCrO ₄	1.2×10^{-10}	AlPO ₄	9.8×10^{-21}	CaSO ₄	4.9×10^{-5}
PbCrO ₄	2.3×10^{-13}	Ca ₃ (PO ₄) ₂	2.1×10^{-33}	PbSO ₄	2.5×10^{-8}
Ag ₂ CrO ₄	1.1×10^{-12}	Mg ₃ (PO ₄) ₂	1.0×10^{-24}	Ag ₂ SO ₄	1.2×10^{-5}

The values in the table can be used to determine the solubility of a sparingly soluble compound.

Let's try to write the equation and K_{sp} for some of them



$$K_{sp} = [\text{Ag}^+]^2 \cdot [\text{SO}_4^{2-}]$$



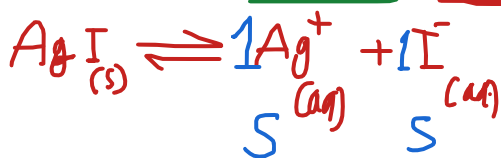
$$K_{sp} = [\text{Ca}^{2+}]^3 \cdot [\text{PO}_4^{3-}]^2$$



$$\text{Solubility} = S$$

Determining solubility (s)

Determine the solubility of AgI in mol/L at 298K, $K_{sp} = 8.5 \times 10^{-17}$



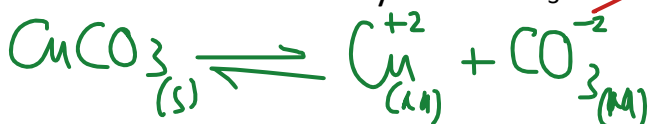
$$K_{sp} = [\text{Ag}^+] \cdot [\text{I}^-]$$

$$8.5 \times 10^{-17} = S \times S$$

$$\Rightarrow \boxed{S = 9.2 \times 10^{-9} \text{ mol/L}}$$

$$\begin{array}{|l} [\text{Ag}^+] = S \\ [\text{I}^-] = S \end{array}$$

Calculate the solubility for CuCO_3 at 298K $K_{sp} = 2.5 \times 10^{-10}$



$$K_{sp} = [\text{Cu}^{+2}] \cdot [\text{CO}_3^{-2}]$$

$$2.5 \times 10^{-10} = S \times S$$

$$\Rightarrow S = 1.58 \times 10^{-5} \text{ mol/L}$$

$$\begin{array}{|l} \text{Solubility} = S \\ [\text{Cu}^{+2}] = S \\ [\text{CO}_3^{-2}] = S \end{array}$$

Calculate the solubility for the following ionic compounds at 298K

a. PbCrO_4

$$K_{sp} = 2.3 \times 10^{-13}$$



$$K_{sp} = [\text{Pb}^{+2}] \cdot [\text{CrO}_4^{-2}]$$

$$2.3 \times 10^{-13} = S \times S$$

$$\boxed{S = 4.79 \times 10^{-7} \text{ mol/L}}$$

$$\begin{array}{|l} \text{Solubility} = S \\ [\text{Pb}^{+2}] = S \\ [\text{CrO}_4^{-2}] = S \end{array}$$

b. CaCO_3

$$K_{sp} = 3.4 \times 10^{-9}$$



$$K_{sp} = [\text{Ca}^{+2}] \cdot [\text{CO}_3^{-2}]$$

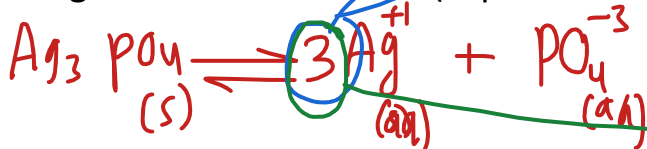
$$3.4 \times 10^{-9} = S \cdot S$$

$$\Rightarrow \boxed{S = 5.83 \times 10^{-5} \text{ mol/L}}$$

$$\begin{array}{|l} \text{Solubility} = S \\ [\text{Ca}^{+2}] = S \\ [\text{CO}_3^{-2}] = S \end{array}$$

c. Ag_3PO_4

$$(K_{sp} = 2.6 \times 10^{-18})$$



$$K_{sp} = [\text{Ag}^+]^3 \cdot [\text{PO}_4^{-3}]$$

$$2.6 \times 10^{-18} = (3S)^3 \cdot (S)$$

$$\boxed{S = 1.76 \times 10^{-5} \text{ mol/L}}$$

$$\begin{array}{|l} \text{Solubility} = S \\ [\text{Ag}^+] = 3S \\ [\text{PO}_4^{-3}] = S \end{array}$$

d. $\text{Ca}_3(\text{PO}_4)_2$

$$(K_{sp} = 2.1 \times 10^{-33})$$

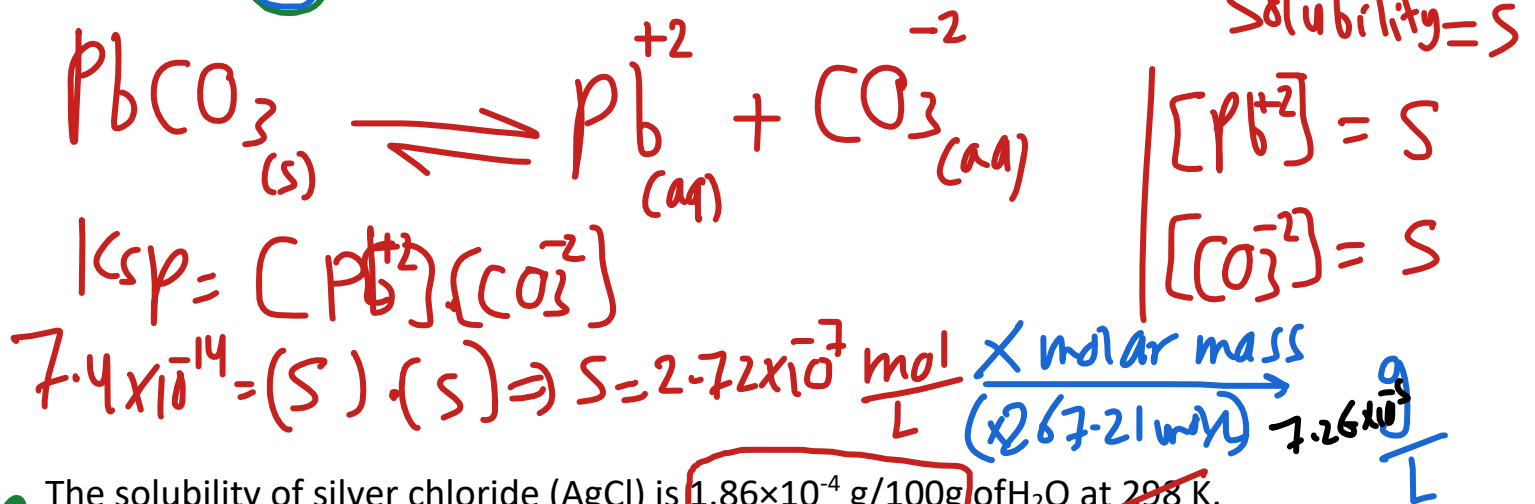


$$K_{sp} = [\text{Ca}^{+2}]^3 \cdot [\text{PO}_4^{-3}]^2$$

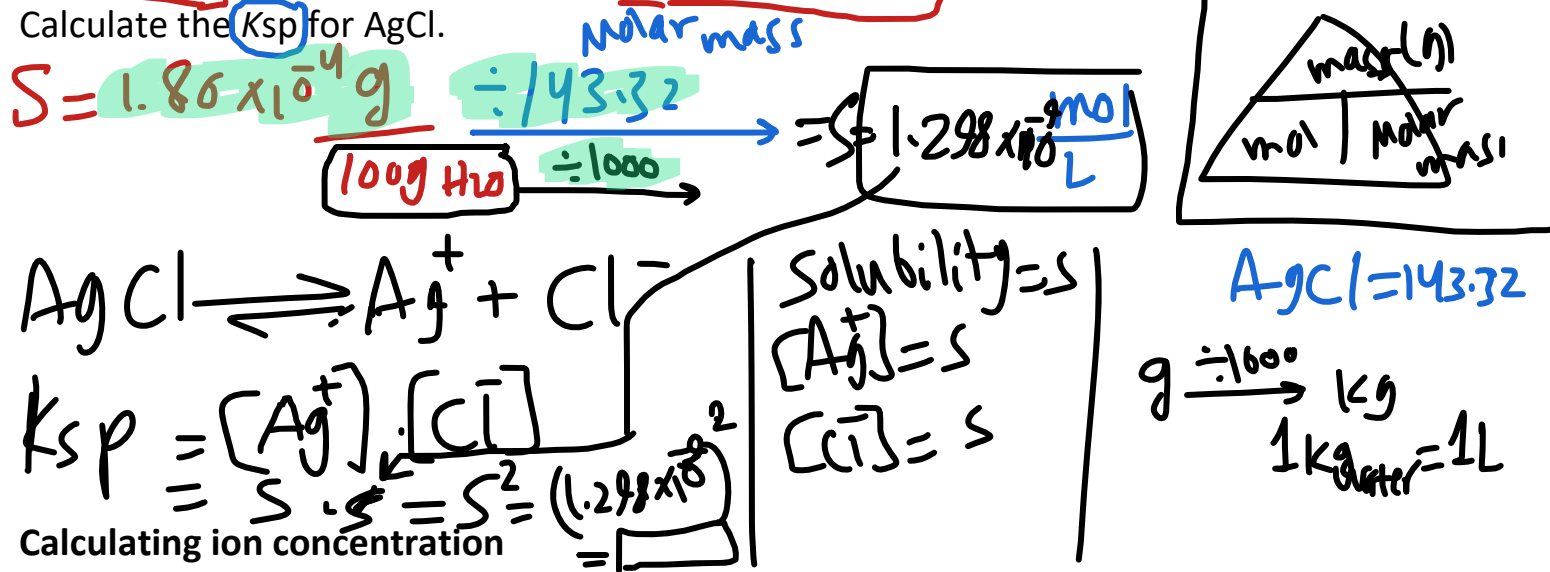
$$2.1 \times 10^{-33} = (3S)^3 \cdot (2S)^2 \Rightarrow \boxed{S = 1.14 \times 10^{-7} \text{ mol/L}}$$

$$\begin{array}{|l} \text{Solubility} = S \\ [\text{Ca}^{+2}] = 3S \\ [\text{PO}_4^{-3}] = 2S \end{array}$$

- The K_{sp} of lead carbonate ($PbCO_3$) is 7.40×10^{-14} at 298 K. What is the solubility of lead carbonate in g/L?



- The solubility of silver chloride ($AgCl$) is 1.86×10^{-4} g/100g of H_2O at 298 K. Calculate the K_{sp} for $AgCl$.



Determine the hydroxide ion concentration in saturated solution of $Mg(OH)_2$ at 298K, $K_{sp} = 5.6 \times 10^{-12}$

Calculate the following.

- a. $[Ag^+]$ in a solution of $AgBr$ at equilibrium $k_{sp} = 5.4 \times 10^{-13}$

b. $[\text{F}^-]$ in a saturated solution of CaF_2

$$K_{\text{sp}} = 3.5 \times 10^{-9}$$

c. $[\text{Ag}^+]$ in a solution of Ag_2CrO_4 at equilibrium $K_{\text{sp}} = 1.1 \times 10^{-12}$

Use the table below to answer the following 3 questions

Concentration Data for the Equilibrium System				
$\text{MnCO}_3(\text{s}) \rightarrow \text{Mn}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$ at 298 K				
Trial	$[\text{Mn}^{2+}]_0$ (M)	$[\text{CO}_3^{2-}]_0$ (M)	$[\text{Mn}^{2+}]_{\text{eq}}$ (M)	$[\text{CO}_3^{2-}]_{\text{eq}}$ (M)
1	0.0000	0.00400	5.60×10^{-9}	4.00×10^{-3}
2	0.0100	0.0000	1.00×10^{-2}	2.24×10^{-9}
3	0.0000	0.0200	1.12×10^{-9}	2.00×10^{-2}

What is the K_{sp} for MnCO_3 at 298 K?

A. 2.24×10^{-11}

C. 1.12×10^{-9}

B. 4.00×10^{-11}

D. 5.60×10^{-9}

What is the molar solubility of MnCO_3 at 298 K?

A. $4.73 \times 10^{-6} \text{ M}$

C. $7.48 \times 10^{-5} \text{ M}$

B. $6.32 \times 10^{-2} \text{ M}$

D. $3.35 \times 10^{-5} \text{ M}$